



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PART TEST

14-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE-S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,D	B,C	A,C	B,C	C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	B	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.05	10.00	92.80	33.30	0.25	4.20	6.00	0.42

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A,B	B,C,D	B,D	B,D	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	B	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	23.00	2.40	8.00	7.00	6.00	14.00	4.00	1.06

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,D	A,B,C,D	A,B	B,C	A,B,D	D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	B	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	84.00	90.00	3.00	987.00	49.00	2.00	2.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)
1. Ans (A,B,D)

The free electrons in metal needs a force to make then accelerate tangentially ($R\alpha$) & radially ($\omega^2 R$) where $\frac{d\omega}{dt} = \alpha$. Now this tangential force may be thought of to be provided by tangential induced electric field such that $eE_{in} = m(R\alpha)$ so induced EMF along circumference = $EMF = 2\pi R \times E_{in}$.

Induced current density $J = \sigma E_{in}$

$$J = \sigma \left(\frac{mR\alpha}{e} \right)$$

In radial direction as ω is not constant so hall voltage will be increasing.

Now magnetic field near center

$$B = \mu_0(ni)$$

where ni = current per unit length of cylinder

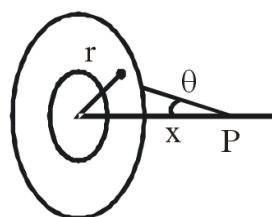
$$= \frac{Jt \cdot \ell}{\ell}$$

$$B = \mu_0 Jt$$

2. Ans (B,C)

For B_p

$$dB = \frac{\mu_0}{4\pi} \frac{2i\pi r^2}{(r^2 + x^2)^{3/2}} \frac{N}{(b-a)} dr$$



$$B_p = \frac{\mu_0}{2} \frac{iN}{(b-a)} \int \frac{r^2}{(r^2 + x^2)^{3/2}} dr$$

Put $r = x \tan \theta$, $dr = x \sec^2 \theta d\theta$

$$\begin{aligned} &= \frac{\mu_0 i N}{2(b-a)} \int \frac{\sec^2 \theta d\theta}{\sec \theta} - \frac{\sec^2 \theta}{\sec^3 \theta} d\theta \\ &= \frac{\mu_0 i N}{2(b-a)} \int_{\pi/6}^{\pi/3} (\sec \theta - \cos \theta) d\theta \\ &= \frac{\mu_0 i N}{2(b-a)} [\ln(\sec \theta + \tan \theta) - \sin \theta]_{\pi/6}^{\pi/3} \\ &= \frac{\mu_0 i N}{2(b-a)} \left[\ln \left(\frac{2+\sqrt{3}}{\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}} \right) - \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right) \right] \\ B_P &= \frac{\mu_0 i N}{2(b-a)} \left[\ln \left(\frac{(2+\sqrt{3})}{\sqrt{3}} \right) - \frac{1}{2}(\sqrt{3}-1) \right] \end{aligned}$$

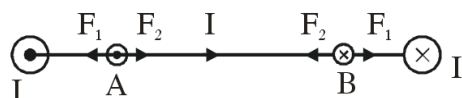
Now $B_O = \int_a^b \frac{\mu_0 i dr N}{2r(b-a)}$

$$B_O = \frac{\mu_0 i N}{2(b-a)} \ln \left(\frac{b}{a} \right) = \frac{\mu_0 i N}{(b-a)} \ln \left(\sqrt{\frac{b}{a}} \right)$$

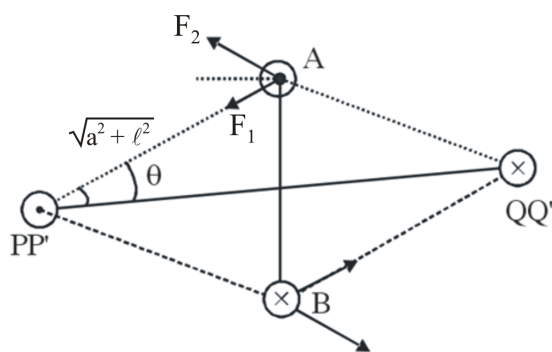
3. Ans (A,C)

Case-(I) $\phi = 0$, as $\vec{\ell}$ is parallel to

$$\vec{F}, \tau = 0$$



Case-(II) $\phi = \frac{\pi}{2}$, $\tau_{\text{net}} \neq 0$



($\theta \rightarrow$ angle as shown)

$$F_1 = F_2 = \frac{\mu_0}{2\pi} \frac{2I^2 \ell}{\sqrt{a^2 + \ell^2}} \&$$

$$\left(\cos \theta = \frac{a}{\sqrt{a^2 + \ell^2}} \right)$$

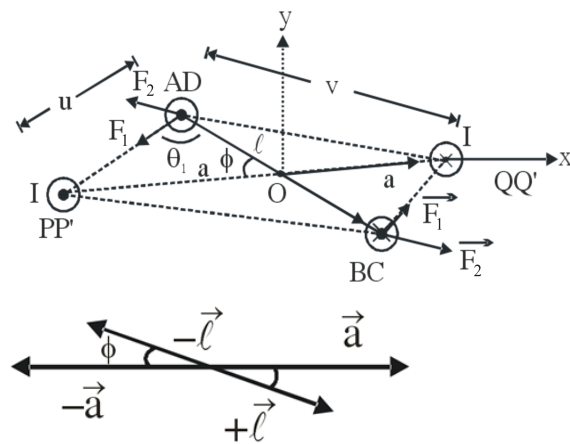
$$\tau_{\text{net}} = [2F_1 \cos \theta \times 2\ell]$$

$$= 4 \frac{\mu_0 I^2 \ell^2 a}{\pi (a^2 + \ell^2)} = \frac{4\mu_0 I^2 \ell^2 a}{\pi (a^2 + \ell^2)}$$

for situation in between these force will higher so torque will be max in between.

Force on wire AB & CD are not going to create torque along z-axis. as force on this segment is along z-axis.

Torque in general situation is



$$|\vec{a} + \vec{\ell}| = v = \sqrt{a^2 + \ell^2 + 2a\ell \cos \theta}$$

$$|\vec{a} - \vec{\ell}| = u = \sqrt{a^2 + \ell^2 - 2a\ell \cos \theta}$$

$$\vec{F}_1 = |\vec{F}_1| \frac{(\vec{a} - \vec{\ell})}{|\vec{a} - \vec{\ell}|}$$

on wire BC, (due to QQ')

$$\vec{\tau}_1 = \vec{\ell} \times \vec{F}_1 = \vec{\ell} \times (\vec{a} - \vec{\ell}) \frac{|\vec{F}_1|}{|\vec{a} - \vec{\ell}|}$$

$$= \frac{\vec{\ell} \times \vec{a} |\vec{F}_1|}{|\vec{a} - \vec{\ell}|} = \frac{\ell a \sin \phi |\vec{F}_1|}{u} \hat{k}$$

on wire BC, (due to PP')

So it's torque on BC

$$\vec{\tau}_2 = \vec{\ell} \times \vec{F}_2 = \vec{\ell} \times \frac{(\vec{a} + \vec{\ell})}{|\vec{a} + \vec{\ell}|} |\vec{F}_2|,$$

$$\vec{\tau}_2 = \frac{\ell a \sin \phi}{v} |\vec{F}_2| \hat{k}$$

Similar forces will act on wire AD on loop.

Now net torque on loop

$$\vec{\tau}_{\text{net}} = 2(\vec{\tau}_1 + \vec{\tau}_2) = 2\ell a \sin \phi \hat{k} \left[\frac{|\vec{F}_1|}{u} + \frac{|\vec{F}_2|}{v} \right]$$

$$|\vec{F}_1| = \frac{\mu_0 I^2}{2\pi u} \cdot 2\ell, \quad |\vec{F}_2| = \frac{\mu_0 I^2}{2\pi v} \cdot 2\ell$$

$$\vec{\tau}_{\text{net}} = 2\ell a \sin \phi \hat{k} \frac{\mu_0 I^2 2\ell}{2\pi} \left[\frac{1}{u^2} + \frac{1}{v^2} \right]$$

$$= \frac{2\mu_0 I^2 \ell^2 a \sin \phi}{\pi} \left[\frac{1}{a^2 + \ell^2 - 2a\ell \cos \phi} + \frac{1}{a^2 + \ell^2 + 2a\ell \cos \phi} \right]$$

$$= \left(\frac{2(a^2 + \ell^2)}{(a^2 + \ell^2)^2 - 4a^2 \ell^2 \cos^2 \phi} \right) \frac{2\mu_0 I^2 \ell^2 a \sin \phi}{\pi}$$

$$= \frac{2\mu_0 I^2 \ell^2 a \sin \phi}{\pi} \hat{k} \left[\frac{2(a^2 + \ell^2)}{(a^2 + \ell^2)^2 - 4a^2 \ell^2 \cos^2 \phi} \right]$$

$$= \frac{4\mu_0 I^2 a \ell^2 \sin \phi}{\pi} \hat{k} \left[\frac{a^2 + \ell^2}{(a^2 - \ell^2)^2 + 4a^2 \ell^2 \sin^2 \phi} \right]$$

4. **Ans (B,C)**

In this case circuit will be in resonance, hence max power will be transferred.

5. **Ans (C,D)**

EMF in body of car will be due to vertical component of earth's magnetic field,

$$B_v = B_H \tan \delta = 10 \text{ Gauss} = 10^{-3} \text{ T}$$

$$\text{EMF} = B_v \ell v = 10^{-3} \times 1.5 \times \left(72 \times \frac{5}{18} \right) = 30 \text{ mV}$$

but same EMF will also induced in bulb connecting metal wires so net EMF in loop is zero & induced current is zero.

6. **Ans (A,B,C,D)**

All points are theoretical. Eddy current stops the coil from oscillating. Soft iron is having high permeability so it increases magnetic field strength. Reading should be taken perpendicularly to avoid parallel error. $i \propto \theta$ so marks are equally spaced.

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (C)**

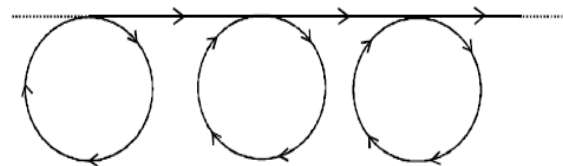
LC circuit with no initial current will have current graph starting from zero and circuit prevents backward flow of current.

8. **Ans (B)**

Induced electric field must be dependent on rate of change of magnetic field, which is directly related with current in coil.

10. **Ans (A)**

$\frac{2\pi m}{eB}$ is time interval of time period of circular motion in magnetic field, so it will follow straight line path for time when magnetic field is off and circular path when magnetic field is on. But distance in straight path $v \times \frac{2\pi m}{eB}$ will be greater than diameter of circular path $2 \frac{mv}{eB}$, so path should be, [in interval of $\frac{4mv}{eB}$]



On an average electron is moving forward by distance of $\left(\frac{2\pi mv}{eB} \right)$ in time interval of $\frac{4\pi m}{eB}$.

so average speed is $\frac{v}{2}$ but $v = \frac{E}{B}$ by velocity selector.

(B) In option 2, time interval is $\frac{\pi m}{2eB}$ so electron will cover quarter arc path when field is switched on then straight line path of $v \times \frac{\pi m}{2eB}$ length so given path is correct.

(C) In option 3, time interval is $\frac{2\pi m}{eB}$ which will make electron to go in full circle when magnetic field is on, then straight line path of $v \left(\frac{2\pi m}{eB} \right)$ length. Given path shown that straight line path is of same length as that of diameter of circle which is incorrect.

(D) In option-4 time interval of $\frac{2\pi m}{3eB}$ will make electron to move in arc of $\frac{2\pi}{3}$ angle, and straight line path. The direction of motion should be clockwise but given is in anticlockwise.

PART-1 : PHYSICS

SECTION-II

1. **Ans (2.05)**

$$I = I_1 + I_2$$

$$I = \frac{100}{50} \sin(\omega t - \phi) + \frac{100}{100} \sin \omega t, (\phi = 37^\circ)$$

$$I = \frac{13}{5} \sin \omega t - \frac{6}{5} \cos \omega t$$

$$I = \frac{\sqrt{205}}{5} \sin(\omega t - 6)$$

$$I_0 = \frac{\sqrt{205}}{5} \text{ A so, } x = 2.05$$

2. **Ans (10.00)**

$$u = \frac{1}{2} LI^2 = \frac{1}{2} LI_0^2 e^{-\frac{2Rt}{L}}$$

$$\text{for } 40\%, e^{-\frac{2Rt}{L}} = 6.4$$

$$t \propto \frac{L}{R}$$

for next 40%, $t = 6 \text{ sec.}$

$$\text{so, } t = 4 + 6 = 10 \text{ sec}$$

3. **Ans (92.80)**

particle moves in a cycloidal path when maximum

$$\text{speed is } \frac{2E}{B}.$$

4. **Ans (33.30)**

$$U_i = -\vec{M} \cdot \vec{B}$$

$$\Rightarrow -[4\hat{i} + 3\hat{j} - 12\hat{k}] \cdot [1.2\hat{i} + 2.5\hat{j} + 3\hat{k}]$$

$$U_i = -[4.8 + 7.5 - 36] = 23.7 \text{ J}$$

$$U_f = -[3\hat{i} + 12\hat{j} - 8\hat{k}] \cdot [1.2\hat{i} + 2.5\hat{j} + 3\hat{k}] = -9.6 \text{ J}$$

$$W_C = -\Delta U$$

$$W_C = -[-9.6 - 23.7] = 33.3 \text{ J}$$

5. **Ans (0.25)**

current in inductor is 5A, so

$$-10 - 20 \frac{dI}{dt} + 20 - 5 = 0$$

$$\Rightarrow \frac{dI}{dt} = 0.25 \text{ A/s}$$

6. **Ans (4.20)**

$$W = -\epsilon \int \vec{E} \cdot d\vec{r}$$

$$\oint \vec{E} \cdot d\vec{r} = \frac{d\phi}{dt}$$

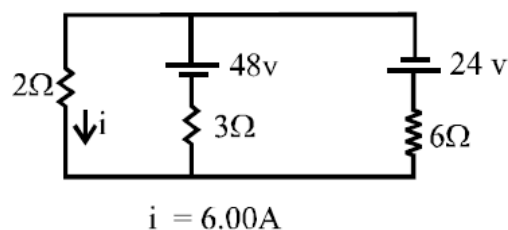
$$\text{from region O12, } \oint \vec{E} \cdot d\vec{r} = 0$$

for radial line, so

$$\oint \vec{E} \cdot d\vec{r} = \frac{d\phi}{dt} = \frac{dB}{dt} \cdot A$$

$$W = \frac{dB}{dt} \cdot A = 4.2 \text{ J}$$

7. **Ans (6.00)**



8. **Ans (0.42)**

$$\vec{P} = \frac{1}{3} P_{\max}$$

$$\left(\frac{E_{\text{rms}}}{Z} \right)^2 R = \frac{1}{3} \frac{E_{\text{rms}}^2}{R}$$

$$\Rightarrow Z^2 = 3R^2 = R^2 + x^2 \Rightarrow x = \sqrt{2}R$$

Now process according to derivation

$$\omega L - \frac{1}{\omega C} = \pm \sqrt{2}R$$

$$LC\omega^2 - (\pm\sqrt{2}RC)\omega - 1 = 0$$

$$\omega = \frac{\pm\sqrt{2}RC \pm \sqrt{4R^2C^2 + 4LC}}{2LC};$$

$$\omega_1 = \frac{+\sqrt{2}RC + \sqrt{4R^2C^2 + 4LC}}{2LC}$$

$$\omega_2 = \frac{-\sqrt{2}RC + \sqrt{4R^2C^2 + 4LC}}{2LC};$$

$$\Delta\omega = \frac{2\sqrt{2}RC}{2LC} = \sqrt{2} \frac{R}{L}$$

so new quality factor is

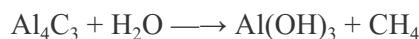
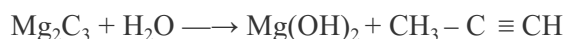
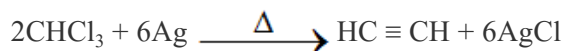
$$Q' = \frac{\omega_0}{\Delta\omega} = \frac{\frac{1}{\sqrt{LC}}}{\sqrt{2}R/L} = \frac{1}{\sqrt{2}} \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{Q}{\sqrt{2}}$$

$$Q' = \frac{0.6}{\sqrt{2}} = 0.42$$

PART-2 : CHEMISTRY

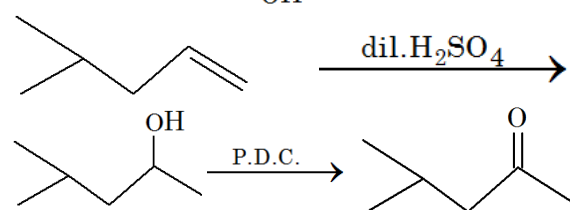
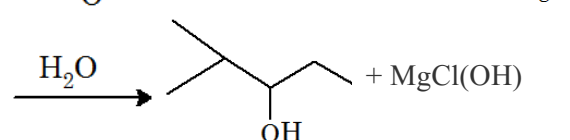
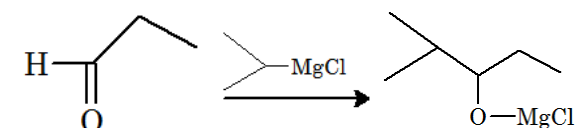
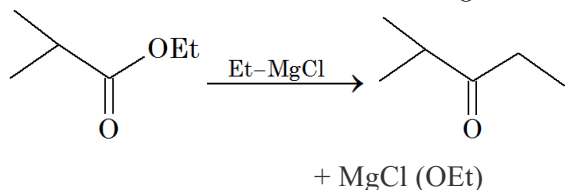
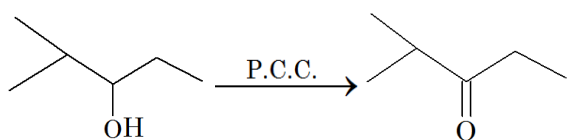
SECTION-I (i)

1. **Ans (A,B,C)**

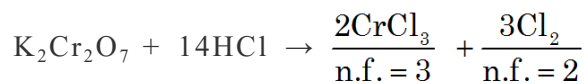


1-Alkyne OR terminal Alkynes give white ppt on reaction with $\text{AgNO}_3 / \text{NH}_4\text{OH}$.

2. **Ans (A,B)**



6. **Ans (B,D)**



$$\frac{0.2 \times V_{\text{K}_2\text{Cr}_2\text{O}_7}}{1} = \frac{70 \times 0.1}{14}$$

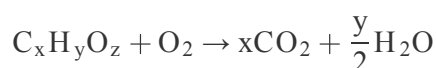
$$\Rightarrow V_{\text{K}_2\text{Cr}_2\text{O}_7} = \frac{70}{28} = 2.5\text{mL}$$

$$\text{Moles of Cl}_2 \text{ formed} = 1 \times 10^{-3} \text{ mole}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (C)**



$$\text{moles of carbon} = \frac{8.8}{44} = 0.2 \text{ mole}$$

$$\Rightarrow 2.4 \text{ gm of carbon}$$

$$\text{moles of H}_2\text{O} = \frac{3.6}{18} = 0.2 \text{ moles}$$

$$\text{mass of hydrogen} = 0.2 \times 2 = 0.4 \text{ gm}$$

$$\text{mass of oxygen in acid} = 6 - 0.4 - 2.4 = 3.2 \text{ gm}$$

$$\text{mole ratio of C : H : O} = 0.2 : 0.4 : 0.2$$

$$= 2 : 4 : 2 = 1 : 2 : 1$$

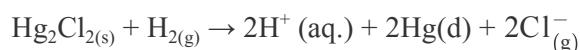
$$\text{Empirical formula of acid} = \text{CH}_2\text{O}$$

$$\text{Molecular formula} = \text{C}_2\text{H}_4\text{O}_2$$

$$\Rightarrow \text{CH}_3\text{COOH}$$

8. **Ans (D)**

$$\text{Conc. of acid} = \frac{6}{60} = 0.1\text{M}$$



$$Q = \frac{[\text{H}^+]^2 [\text{Cl}^-]^2}{(\text{pH}_2)} = [\text{H}^+]^2$$

$$0.46 = (0.28 - 0) - \frac{0.06}{2} \log[\text{H}^+]$$

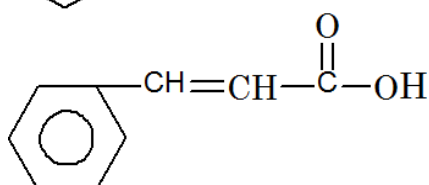
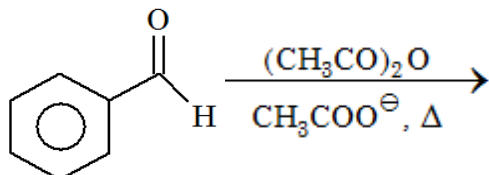
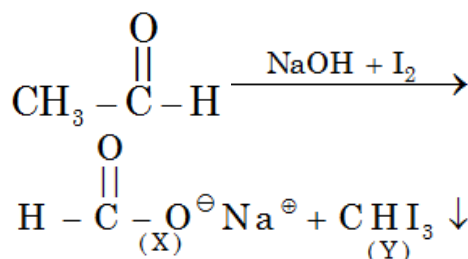
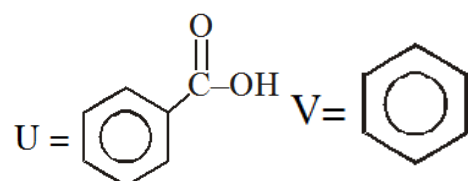
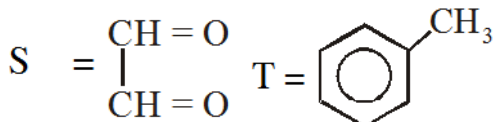
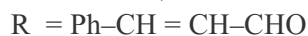
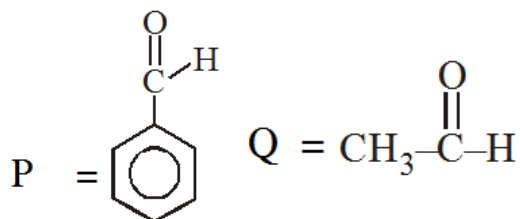
$$\Rightarrow 0.18 = -0.06 \log[\text{H}^+]$$

$$\Rightarrow \log[\text{H}^+] = -3 \Rightarrow [\text{H}^+] = 10^{-3}\text{M} = C\alpha$$

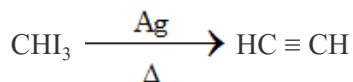
$$\Rightarrow \alpha = 10^{-2}$$

$$K_a \approx C\alpha^2 = 10^{-5}$$

10. Ans (B)



Cinnamic acid



PART-2 : CHEMISTRY

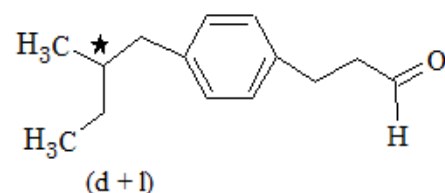
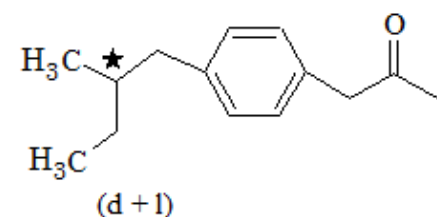
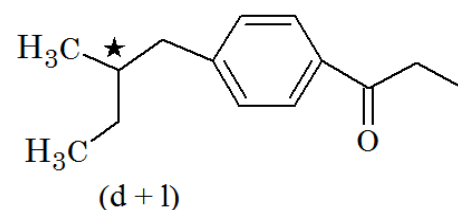
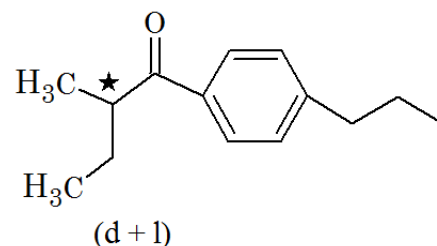
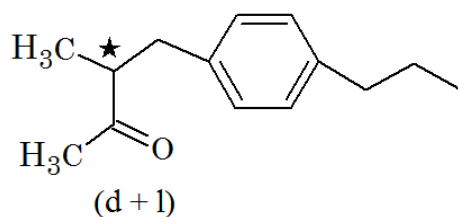
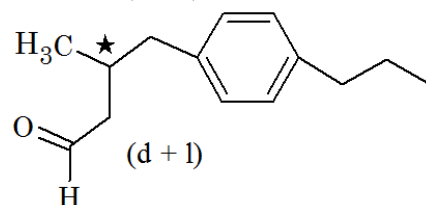
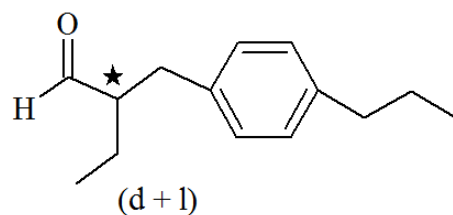
SECTION-II

5. Ans (6.00)

(II) , (III) , (IV) , (V) , (VI) , (X)

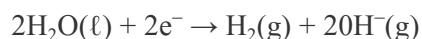
Aldehyde hemiacetal & HCOOH can give
tollen's test.

6. Ans (14.00)



7. Ans (4.00)

At cathode



8. Ans (1.06)

Loss in wt. due to heating

$$= 6.2 \times \frac{5}{100} = 0.31 \text{ g}$$



$$2 \times 84 \text{ gm} \quad 44 + 18 \text{ gm loss}$$

1 mole $\text{NaHCO}_3 \rightarrow 31 \text{ gm loss due to heating}$

0.31 gm loss from 0.01 mole of NaHCO_3

$$\text{Moles of } \text{Na}_2\text{CO}_3 \text{ produced} = \frac{0.01}{2} = 0.005$$

Total moles of carbonate reacted with BaCl_2

$$= \left(7.5 \times 0.2 \times \frac{100}{10} \right) \times 10^{-3} = 0.15$$

Moles of carbonates in original sample

$$= 0.15 - 0.005 = 0.01$$

Mass of Na_2CO_3 in original sample

$$= 0.01 \times 106 = 1.06 \text{ gram}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,D)

$$\text{Let } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

be an ellipse

$$\text{Area} = A_1 = \pi ab$$

Let (h, k) be the mid-point of PS

$$\Rightarrow 2h = a \cos \theta + ae \text{ \& } 2k = b \sin \theta$$

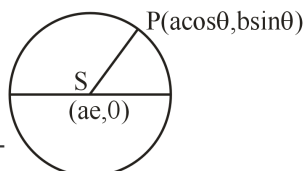
Eliminating θ we get

$$\frac{\left(x - \frac{ae}{2}\right)^2}{\left(\frac{a}{2}\right)^2} + \frac{y^2}{\left(\frac{b}{2}\right)^2} = 1$$

$$A_2 = \pi \frac{a}{2} \frac{b}{2} = \frac{\pi ab}{4}$$

$$\Rightarrow A_1 : A_2 = 4 : 1$$

$$\frac{x}{y} = 4$$



2. Ans (A,B,C,D)

$$\frac{\log_{3^4} 3^6 + \log_{3^6} 3^4}{\log_{\left(\frac{9}{4}\right)^{1/2}} \frac{9}{4} + \log_{\frac{9}{4}} \left(\frac{9}{4}\right)^{1/2}} = \frac{\frac{3}{2} + \frac{2}{3}}{2 + \frac{1}{2}} = \frac{13}{15}$$

$$\therefore a = 13, b = 15$$

$$\therefore b - a = 2.$$

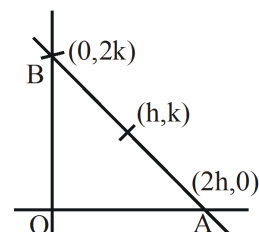
3. Ans (A,B)

Area of triangle OAB

$$= \frac{1}{2} \cdot 2h \cdot 2k = \pm 4$$

$\Rightarrow$ locus of middle point

$$\text{of AB is } |xy| = 2$$

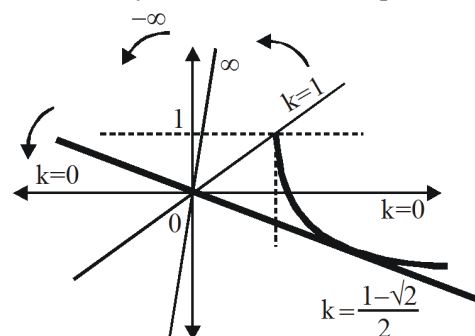


4. Ans (B,C)

Given curve is lower branch of parabola

$$(y - 1)^2 = x - 1$$

and line is $y = kx$ where k is slope of line



5. Ans (A,B,D)

$$y = \frac{\sqrt{2} \sin x}{3 - \cos x}$$

$$= \frac{\sqrt{2} \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}}{\frac{3(1 + \tan^2 \frac{x}{2}) - (1 - \tan^2 \frac{x}{2})}{1 + \tan^2 \frac{x}{2}}} = \frac{2\sqrt{2}t}{2 + 4t^2}$$

$$y = \frac{\sqrt{2}t}{1 + 2t^2} \Rightarrow 2yt^2 - \sqrt{2}t + y = 0$$

$$t \in \mathbb{R} \Rightarrow D > 0$$

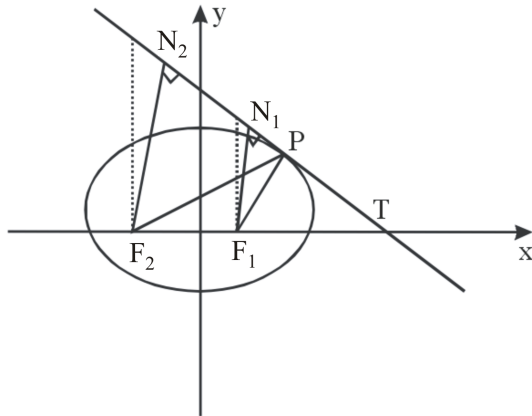
$$2 - 8y^2 \geq 0$$

$$4y^2 - 1 \leq 0 \quad y \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$$n = -\frac{1}{2}, m = \frac{1}{2}$$

6. Ans (D)

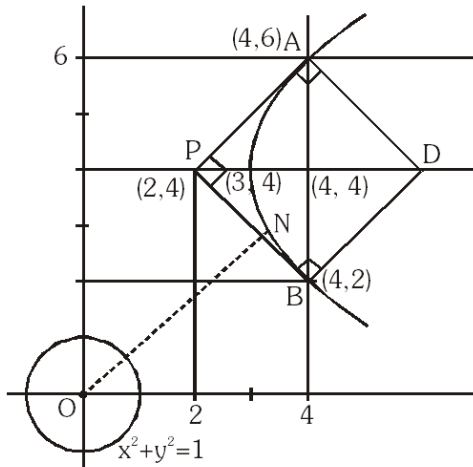
$$\begin{aligned}
 & (\text{Area of } \triangle PTF_1) \cdot (\text{Area of } \triangle PTF_2) \\
 &= \left(\frac{1}{2} PT \cdot F_1 N_1 \right) \left(\frac{1}{2} PT \cdot F_2 N_2 \right) \\
 &= \frac{PT^2 \cdot b^2}{4} \\
 &\sum_{i=1}^8 \frac{(\text{area of } \triangle P_i T_i F_1) (\text{area of } \triangle P_i T_i F_2)}{(P_i T_i)^2} = \frac{1}{4} \times 9 \times 8 \\
 &= 18 \text{ square units}
 \end{aligned}$$



PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (C)

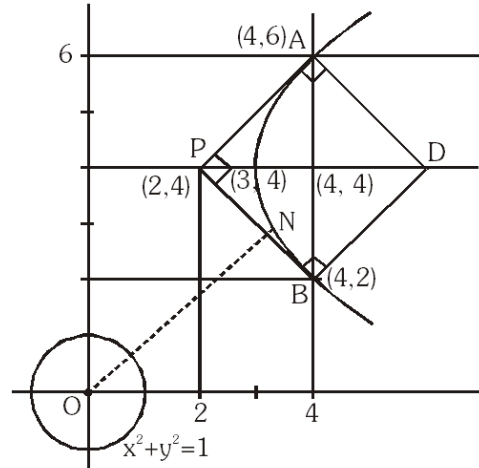


The directrix must touch the circle $x^2 + y^2 = 2r^2$

$$\therefore \sqrt{2}r = 2$$

$$r = \sqrt{2}$$

8. Ans (B)



Let ON be the minimum distance line of parabola from circle

$$N \equiv (x_1, y_1) \equiv (t^2 + 3, 2t + 4)$$

ON must be normal to circle as well as parabola.

$$\frac{2t + 4}{t^2 + 3} = -t$$

$$2t + 4 = -t^3 - 3t$$

$$t^3 + 5t + 4 = 0$$

$$(x_1 + 2)(y_1 - 4) = (t^2 + 5)2t = 2(t^3 + 5t) = -8$$

9. Ans (A)

$$\begin{aligned}
 a^2 + b^2 &= \left(\sin \theta - \sin \frac{\theta}{2} \right)^2 + \left(\cos \theta + \cos \frac{\theta}{2} \right)^2 \\
 &= 2 + 2 \left(\cos \theta \cos \frac{\theta}{2} - \sin \theta \sin \frac{\theta}{2} \right) \\
 &= 2 + 2 \cos \left(\frac{3\theta}{2} \right) \\
 a^2 + b^2 &= 2 + 2 \cos \left(\frac{3 \times 80^\circ}{2} \right) = 1
 \end{aligned}$$

10. Ans (A)

$$\begin{aligned}
 a &= 2 \sin \frac{\theta}{4} \cos \frac{3\theta}{4} \text{ \& } b = 2 \cos \frac{3\theta}{4} \cos \frac{\theta}{4} \\
 b^2 - a^2 &= 4 \cos^2 \frac{3\theta}{4} \left(\cos^2 \frac{\theta}{4} - \sin^2 \frac{\theta}{4} \right) \\
 &= 4 \cos^2 \frac{3\theta}{4} \cos \frac{\theta}{2}
 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1.00)**

$$t_1 = \sqrt{30 + \sqrt{30 + \sqrt{30}}} \Rightarrow t_1 = \sqrt{30 + t_1}$$

$$\Rightarrow t_1^2 - t_1 - 30 = 0$$

$$t_1 = 6$$

$$t_2 = \sqrt{2 + \sqrt{2 + \dots}}$$

$$\Rightarrow t_2 = \sqrt{2 + t_2} = t_2^2 = 2 + t_2$$

$$t_2^2 - t_2 - 2 = 0 \Rightarrow t_2 = 2$$

$$x = \log_3 \left(\frac{6}{2} \right) = 1$$

2. **Ans (84.00)**

Any point on hyperbola $xy = 1$ is $(t, 1/t)$.

Now image of $(t, 1/t)$ by the line mirror

$$y = 2x \text{ is } \frac{x-t}{2} = \frac{y-1/t}{-1} = -2 \left(\frac{2t-1/t}{5} \right)$$

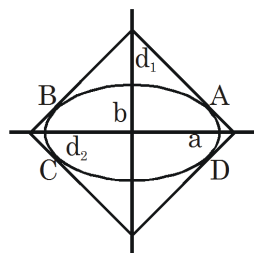
$$\Rightarrow 5x = \frac{4}{t} - 3t \text{ \& } 5y = 4t + \frac{3}{t}$$

Now eliminating t from above two equations we get,

$$12x^2 - 12y^2 - 7xy + 25 = 0$$

$$\therefore b.c = (-7)(-12) = 84$$

3. **Ans (90.00)**



$$d_1 = d_2 = d = \sqrt{a^2 + b^2}$$

Point of intersection of tangents lies on director circle. Hence area

$$= 4 \times \frac{1}{2} (d)^2 = 2 (a^2 + b^2) = 2(45)$$

4. **Ans (3.00)**

Given

$$2 \left(\sin \frac{A+B}{2} \cos \frac{A-B}{2} + \sin \frac{C+D}{2} \cos \frac{C-D}{2} \right) = 2 \times 2$$

$$\Rightarrow \sin A + \sin B + \sin C + \sin D = 4$$

$$\Rightarrow \sin A = \sin B = \sin C = \sin D = 1$$

$$\Rightarrow \angle A = \angle B = \angle C = \angle D = 90^\circ$$

$$\therefore \sum \cos \frac{A}{2} \cos \frac{B}{2} = \sum \cos 45^\circ \cos 45^\circ = 6 \cdot \left(\frac{1}{\sqrt{2}} \right)^2 = 3$$

5. **Ans (987.00)**

$$x^2 - x - 1 = 0$$

$$\Rightarrow x - \frac{1}{x} = 1 \Rightarrow x^2 + \frac{1}{x^2} = 3$$

$$\Rightarrow x^4 + \frac{1}{x^4} = 7 \Rightarrow x^8 + \frac{1}{x^8} = 47$$

$$\log_{10}(x^4 + 1)(x^8 + 1)(x^{16} + 1) - 14 \log_{10}(-x)$$

$$= \log_{10} x^{14} \left(x^2 + \frac{1}{x^2} \right) \left(x^4 + \frac{1}{x^4} \right) \left(x^8 + \frac{1}{x^8} \right) - \log_{10} x^{14}$$

$$= \log_{10} \left(x^2 + \frac{1}{x^2} \right) \left(x^4 + \frac{1}{x^4} \right) \left(x^8 + \frac{1}{x^8} \right)$$

$$= \log_{10} 3 \times 7 \times 47$$

$$= \log_{10} 987$$

6. **Ans (49.00)**

F is focus and let $RF = \ell$

and $PF = 2RF$

in ΔOPF ,

$$PF^2 + OF^2 = 25$$

$$\Rightarrow 4\ell^2 + (5 - \ell)^2 = 25$$

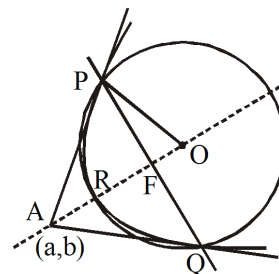
$$\Rightarrow \ell = 2$$

$\therefore$ length of latus

rectum = 8

$$\therefore A \left(\frac{-7}{\sqrt{2}}, \frac{-7}{\sqrt{2}} \right)$$

$$\Rightarrow a^2 + b^2 = \frac{49}{2} + \frac{49}{2} = 49$$



7. **Ans (2.00)**

Let middle pt of chord (h, k)

$$hx - ky = h^2 - k^2$$

$$\Rightarrow y = \frac{h}{k}x + \left(\frac{k^2 - h^2}{k} \right)$$

it is tangent to $y^2 = 12x$ so

$$\frac{k^2 - h^2}{k} = \frac{3k}{h} \Rightarrow h^3 - hk^2 + 3hk = 0$$

$$\text{Locus is } x^3 - xy^2 + 3xy = 0$$

8. **Ans (2.00)**

Equation of chord with given middle point (x, y) is

$$T = S_1 \Rightarrow \frac{xx_1}{4} + \frac{yy_1}{9} = \frac{x_1^2}{4} + \frac{y_1^2}{9} \dots (1)$$

Compare (1) with $9x - 4y = 13$

$$x_1 = 1 \text{ and } y_1 = -1 \text{ (i.e. } \alpha = 1 \text{ \& } \beta = -1)$$

$$CP^2 = \sqrt{(1-0)^2 + (-1-0)^2}$$

$$CP^2 = 2$$

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE-S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	D	B	A	A	C	D	A	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	D	A	D	C	C	A	D	A

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	C	B	A	C	A	C	B	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	B	C	C	C	C	C	C	C

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	C	D	C	A	C	C	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	A	A	C	A	D	B	B	B	A

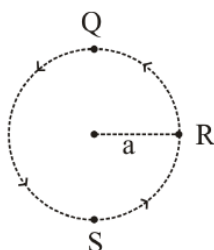
HINT – SHEET
PART-1 : PHYSICS
SECTION-I

 1. **Ans (C)**

$$\vec{B} = \vec{B}_{\text{Curved}} + \vec{B}_{\text{Straight}} = \frac{1}{4} \cdot \frac{\mu_0 i}{2R} \otimes + \frac{\mu_0 i}{4\pi \left(\frac{R}{\sqrt{2}}\right)} (2 \sin 45^\circ) \odot$$

$$= \frac{\mu_0 i}{8R} \otimes + \frac{\mu_0 i}{2\pi R} \odot$$

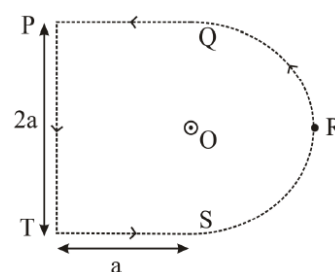
$$= \frac{\mu_0 i}{R} \left[\frac{1}{2\pi} - \frac{1}{8} \right] \odot$$

 2. **Ans (A)**


For the circular Amperian loop

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i \Rightarrow 2 \int_{SRQ} \vec{B} \cdot d\vec{\ell} = \mu_0 i$$

$$\therefore \int_{SRQ} \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 i}{2}$$



For given amperian loop

$$\oint \vec{B} \cdot d\vec{\ell} = \int_{SRQ} \vec{B} \cdot d\vec{\ell} + \int_{QPTS} \vec{B} \cdot d\vec{\ell}$$

$$\mu_0 i = \frac{\mu_0 i}{2} + \int_{QPTS} \vec{B} \cdot d\vec{\ell}$$

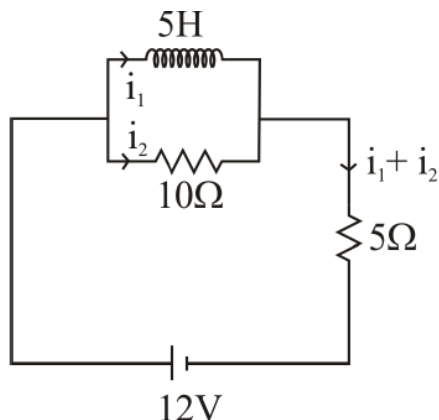
$$\int_{QPTS} \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 i}{2}$$

3. Ans (D)

$$\frac{5di_1}{dt} = 10i_2 \quad \dots(i)$$

$$12 - 10i_2 - 5(i_1 + i_2) = 0$$

$$12 = 5i_1 + 15i_2 \quad \dots(ii)$$



Just after closing the switch $i_1 \simeq 0$.

$$\therefore i_2 = \frac{12}{15} \text{ A (Put } i_1 = 0 \text{ in eq(ii))}. \quad \text{Option-C}$$

For $t = 0$, EMF induced across inductor decreases with time.

$$\text{EMF} = \frac{5di_1}{dt} = 10i_2 \quad (\text{Both decrease with time})$$

$\therefore i_2$ decrease with time and finally becomes zero.

Option-A

At steady state, $i_2 \rightarrow 0$ (Discussed above)

$$\therefore \text{Current through battery} = i_1 + i_2 = i_1 = \frac{12}{5} \text{ A}$$

(Put $i_2 = 0$ in eq(ii))

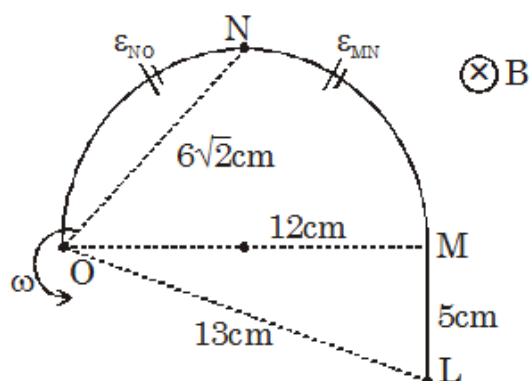
4. Ans (B)

$$\epsilon_{\text{curved segment NO}} = \epsilon_{\text{straight segment NO}} = \frac{1}{2}B\omega(6\sqrt{2})^2$$

$$\epsilon_{MN} + \epsilon_{NO} = \epsilon_{\text{Induced in straight segment OM}}$$

$$= \frac{1}{2}B\omega(12)^2$$

$$\therefore \epsilon_{MN} + 36B\omega = 72B\omega \Rightarrow \epsilon_{MN} = 36B\omega$$



$$\text{Similarly, } \epsilon_{LM} + \epsilon_{MNO} = \epsilon_{\text{straight segment OL}}$$

$$= \frac{1}{2}B\omega(13)^2$$

$$\therefore \epsilon_{LM} = \frac{169B\omega}{2} - 72B\omega = \frac{25B\omega}{2}$$

$$\therefore \text{Ratio} = \frac{\epsilon_{MN}}{\epsilon_{LM}} = \frac{36B\omega}{\left(\frac{25B\omega}{2}\right)} = \frac{72}{25}$$

5. Ans (A)

It is a capacitive circuit. So, current leads voltage

$$\text{by } \phi = \cos^{-1} \left(\frac{R}{Z} \right) = \cos^{-1} \left(\frac{8}{10} \right) = 37^\circ$$

$$i_{\text{peak}} = \frac{V_{\text{peak}}}{Z} = \frac{30}{10} = 3 \text{ A}$$

$$\therefore i = 3 \cos(50\pi t + 37^\circ) \quad (\text{Current in circuit})$$

$$V_R = iR = 24 \cos(50\pi t + 37^\circ) = 24 \cos(50\pi t + 37^\circ)$$

$$V_L = 3 \times 2 \cos(50\pi t + 37^\circ + 90^\circ) = 6 \cos(50\pi t + 127^\circ)$$

$$V_C = 8 \times 3 \cos(50\pi t + 37^\circ - 90^\circ) = 24 \cos(50\pi t - 53^\circ)$$

$$\text{Applied voltage} \Rightarrow 18V = 30 \cos(50\pi t)$$

$$\Rightarrow \cos 50\pi t = \frac{18}{30} = \frac{3}{5} \Rightarrow 50\pi t = 53^\circ$$

$$\therefore V_L = 6 \cos(53^\circ + 127^\circ) = -6V$$

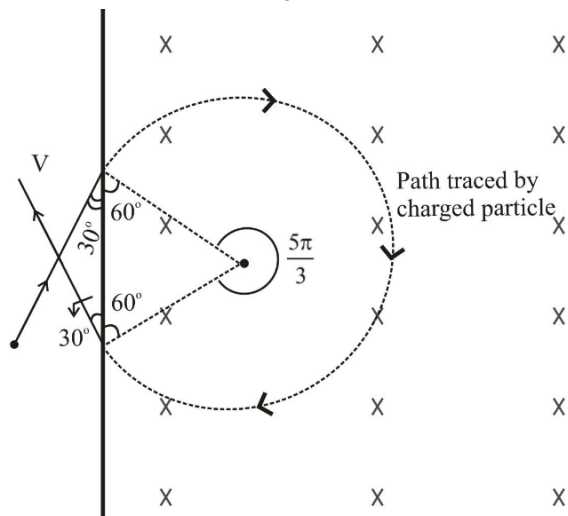
$$V_C = 24 \cos(53^\circ - 53^\circ) = 24V$$

$$V_R = 24 \cos(53^\circ + 37^\circ) = 0$$

$$\therefore V_A - V_B = V_L + V_C = 18V$$

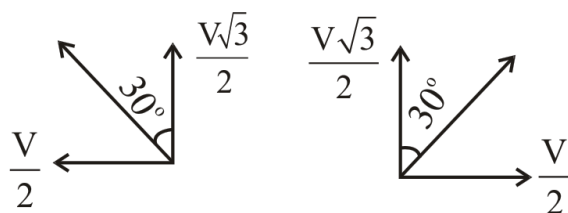
6. **Ans (A)**

$$\vec{F}_{\text{avg}} = \frac{\text{Change in Momentum}}{\text{Time}}$$



$$\text{Time spent} = \frac{5\pi}{3} \cdot T = \frac{5}{3} \times \frac{\pi m}{qB}$$

$$\text{Change in momentum} = -2mv \sin 30^\circ = -mv \hat{i}$$



$$\therefore |\vec{F}_{\text{avg}}| = \frac{mv}{\left(\frac{5\pi m}{3qB}\right)} = \frac{3qBV}{5\pi}$$

7. **Ans (C)**

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{L \times 20 \times 10^{-6}}} = \frac{10^4}{8\pi} \text{ Hz}$$

$$\Rightarrow L = \frac{16}{20} \times 10^{-2} \text{ H} = 8 \text{ mH}$$

$$\begin{aligned} \text{So, total energy of LC circuit} &= \frac{1}{2} Li^2 + \frac{q^2}{2C} \\ &= \frac{1}{2} \times 8 \times 10^{-3} \times 2^2 + \frac{(0.6 \times 10^{-3})^2}{2 \times 20 \times 10^{-6}} \text{ Joules} \\ &= 16 \text{ mJ} + 9 \text{ mJ} = 25 \text{ mJ} \end{aligned}$$

If Q_0 = Max charge on capacitor

$$\frac{Q_0^2}{2C} = 25 \text{ mJ} \Rightarrow Q_0^2 = 25 \times 10^{-3} \times 2 \times 20 \times 10^{-6}$$

$$\Rightarrow Q_0 = 1 \text{ mC}$$

$$\therefore q(t) = Q_0 \sin(\omega t + \phi)$$

$$= 1 \text{ mC} \sin(2500t + \phi)$$

$$\left(\omega = \frac{1}{\sqrt{LC}} = 2\pi f = \frac{10^4}{4} \text{ rad/sec} = 2500 \text{ rad/sec} \right)$$

$$\text{At } t = 0, q = 0.6 \text{ mC} \Rightarrow 0.6 \text{ mC} = 1 \text{ mC} \sin \phi$$

$$\Rightarrow \sin \phi = \frac{3}{5} \Rightarrow \phi = 37^\circ \text{ or } 143^\circ$$

$\phi = 143^\circ$ is correct value of initial phase angle as charge on right plate will decrease initially due to outgoing current.

$$\therefore q(t) = 1 \text{ mC} \sin(2500t + 143^\circ)$$

$$= 1 \text{ mC} \cos(2500t + 143^\circ) \text{ Option: (C)}$$

8. **Ans (D)**

In transformers, moving coil galvanometers etc., soft iron core is used as they could be easily magnetized, demagnetized and hysteresis loss too are less.

9. **Ans (A)**

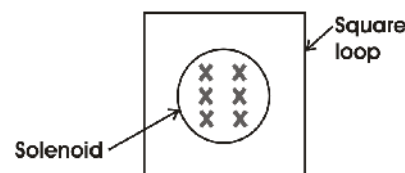
n = Number of turns p.u length

$$= \frac{1 \text{ meter}}{0.5 \text{ mm}} = 2000 \text{ turns/meter}$$

$$\therefore B_{\text{Solenoid}} = \mu_r \mu_0 n i = 200 \times 4\pi \times 10^{-7} \times 2000 \times 2 \text{ Tesla}$$

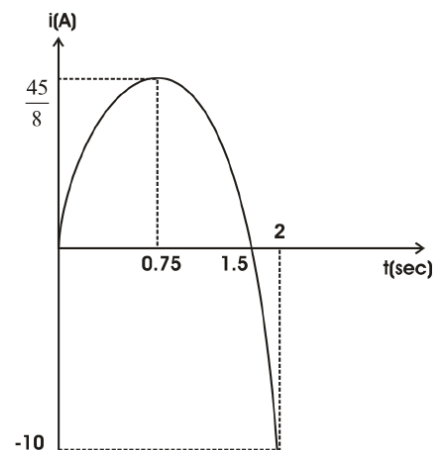
$$= 32\pi \times 10^{-2} \text{ Tesla} = \frac{8\pi}{25} \text{ Tesla}$$

10. **Ans (C)**



$$\phi_S = (\mu_0 n i(t)) (\pi R^2)$$

$$E_S = \frac{-d\phi_S}{dt} = -[\pi R^2 \mu_0 n] \times \frac{di(t)}{dt}$$



$$\frac{di(t)}{dt} \neq 0 \text{ at } t = 0 \text{ and } t = 1.5 \text{ sec. Option A}$$

is incorrect.

$$\phi(t = 2 \text{ sec}) > \phi(t = 0.75 \text{ sec}) \quad \text{Option B}$$

is incorrect.

$$\phi(t = 1.5 \text{ sec}) = 0 \quad \therefore i(t) = 0$$

Also, induced current changes its direction at $t = 0.75 \text{ sec}$.

11. **Ans (C)**

$$P_{\max} = \frac{V_o^2}{2R} = \frac{(200\sqrt{2})^2}{200} = 400W$$

12. **Ans (C)**

Let new magnetic moment in first case is m_1 and in 2nd case m_2

$$\text{Then } 2m_1 = 2m_2 = M \Rightarrow M_1 = M_2 = \frac{M}{2}$$

$$\text{Now } M_1 = m_1 l \text{ and } M_2 = m_2 \frac{l}{2}$$

Where m_1 and m_2 are new pole strength

$$\Rightarrow m_1 = \frac{M}{2l}, \quad m_2 = \frac{m}{l}, \quad (\because M = ml)$$

$$\Rightarrow m_1 = \frac{m}{2}, \quad m_2 = m$$

14. **Ans (A)**

For critical damping $\omega_0 = \gamma$

$$\frac{1}{\sqrt{LC}} = \frac{R}{2L}$$

$$\frac{4L^2}{LC} = R^2 \Rightarrow 4L = R^2 C$$

$$R = 2\sqrt{\frac{L}{C}}$$

$$R = 2\sqrt{\frac{4}{1 \times 10^{-6}}} = 2 \times 10^3 \Omega$$

15. **Ans (D)**

$$E_B = 3 E_C$$

$$\Rightarrow \frac{1}{2} Li^2 = 3 \frac{1}{2} \frac{Q^2}{C} \Rightarrow 2^{-2} = \frac{3Q^2}{LC}$$

$$i = w\sqrt{Q_0^2 - Q^2} \Rightarrow \cancel{w}^2 (Q_0^2 - Q^2) = \frac{3Q^2}{LC}$$

$$\Rightarrow Q = +\frac{Q_0}{2}$$

$$\pm \frac{Q_0}{2} = Q_0 \sin \left(\cot + \frac{\pi}{3} \right)$$

$$\Rightarrow \cot + \frac{\pi}{3} = \frac{5\pi}{6} \text{ or } \frac{7\pi}{6}$$

$$\Rightarrow t_{\min} \frac{T}{3} = 6 \text{ sec}$$

16. **Ans (C)**

$$I_o = \frac{V_o}{XL} = \frac{V_o}{WL} \Rightarrow i\alpha \frac{1}{w}$$

17. **Ans (C)**

$$V_R^2 + (V_L - V_C)^2 = V_o^2 \text{ at resonance}$$

$$V_L = V_C$$

$$\Rightarrow V_R = V_o$$

18. **Ans (A)**

$$I_L = \frac{V_o}{X_L}, \quad I_C = \frac{V_o}{X_C} \Rightarrow I_{\text{Net}} = |I_C - I_L|$$

$$= \left(2 \frac{V_o}{R} - \frac{V_o}{R} \right) = \frac{V_o}{R}$$

19. **Ans (D)**

For some charge and mass distribution

$$\frac{M}{L} = \frac{Q}{2m}$$

L = angle moment of H.S. = $I\omega$

$$M = \frac{Q}{2m} L \quad \left\{ I = \frac{2}{5} mR^2 \right\}$$

$$L = \frac{2}{5} mR^2 \omega$$

$$\Rightarrow M \left(\frac{Q}{2m} \right) \cdot \frac{2}{5} mR^2 \omega = \frac{Q\omega R^2}{5}$$

20. **Ans (A)**

$$B = \frac{\mu_0 i}{4\pi \left(\frac{a}{2} \right)} (2) \frac{1}{\sqrt{2}} \times \cancel{A} = \frac{\mu_0 i}{\pi a} 2\sqrt{2}$$

$$Q = B(b^2)$$

$$\frac{Q}{i} = M = \frac{\mu_0 2\sqrt{2}}{\pi a} \cdot b^2$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (1.50)**

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 25 \times 10^{-6}} = 400 \Omega$$

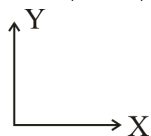
$$\therefore Z = \sqrt{R^2 + X_C^2} = \sqrt{300^2 + 400^2} = 500 \Omega$$

$$\Rightarrow i_{\text{RMS}} = \frac{V_{\text{RMS}}}{Z} = \frac{50(\sqrt{2})}{500} = \frac{1}{10\sqrt{2}} A$$

$$\therefore P_{\text{avg}} = i_{\text{RMS}}^2 R$$

$$= \left(\frac{1}{10\sqrt{2}} \right)^2 \times 300 = 1.5 \text{ watt}$$

2. **Ans (1.41)**



Induced electric field at point 'P' = $\frac{r_1}{2} \frac{dB}{dt}$

$$\therefore \vec{a}_P = \frac{\frac{eR}{2} \left(\frac{dB}{dt} \right)}{m} = \frac{\frac{eR}{4} \frac{dB}{dt}}{2m} = \frac{eR}{8m} \frac{dB}{dt} (-\hat{i})$$

Induced Electric field at point 'S' = $\frac{R^2}{2r_2} \frac{dB}{dt}$

$$\vec{a}_D = \frac{\frac{eR^2}{2r_2} \left(\frac{dB}{dt} \right) \hat{j}}{2m} = \frac{\frac{eR^2}{2(2R)} \left(\frac{dB}{dt} \right) \hat{j}}{2m} = \frac{eR}{8m} \left(\frac{dB}{dt} \right) \hat{j}$$

$$\therefore \frac{|\vec{a}_P + \vec{a}_D|}{|\vec{a}_P|} = \frac{\frac{eR}{8m} \frac{dB}{dt} |-\hat{i} + \hat{j}|}{\frac{eR}{8m} \frac{dB}{dt} |-\hat{i}|} = \sqrt{2} = 1.4142$$

$$= 1.41$$

3. **Ans (98.12)**

$$\Phi_{\text{square frame}} = \frac{B \pi R^2}{2}$$

EMF induced in frame

$$= \frac{\pi R^2}{2} \frac{dB}{dt} = \frac{\pi}{2} \cdot \left(\frac{1}{2} \right)^2 (1 \text{ T/sec}) = \frac{\pi}{8} \text{ volts}$$

Current induced in frame

$$= \frac{\text{EMF}}{\text{Resistance}} = \frac{\pi/8}{4} = \frac{\pi}{32} \text{ A}$$

$$= \frac{3.14}{32} \text{ A}$$

$$= 98.12 \text{ mA}$$

4. **Ans (1.30)**

Magnetic field strength

$$H = ni = \left(\frac{20}{0.5} \right) \times 1 = 40 \text{ A/m}$$

From the graph the magnetic flux density at $H = 40 \text{ A/m}$ is 1.3 T for alloy bar.

$$= 1.3 \text{ T}$$

5. **Ans (32.00)**

Magnetic moment of solenoid per unit length = iNA

(Here, N = Number of turns per unit length)

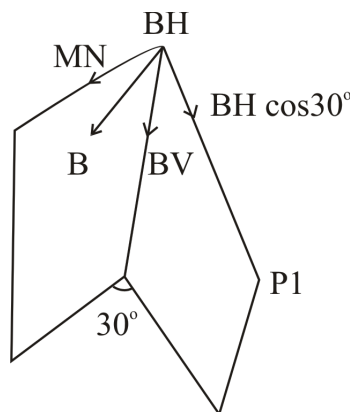
$$\therefore \text{Magnetization vector } \vec{M} = \frac{(iNA) L}{LA}$$

L = Length of Solenoid

$$= iN = (8A)(400 \text{ m}^{-1})$$

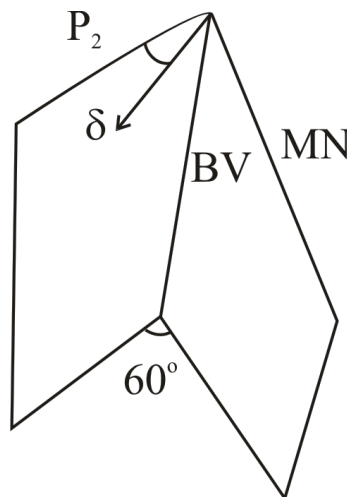
$$= 3200 \text{ Am}^{-1}$$

7. **Ans (3.00)**



$$\frac{BV}{BH \cos 30^\circ} = \tan 60^\circ$$

$$\Rightarrow \frac{BV}{BH} = \sqrt{3} \cdot \frac{\sqrt{3}}{2} = \frac{3}{2}$$



$$\tan \delta = \frac{BV}{BH \cos 60^\circ} = \frac{3}{2 \times \frac{1}{2}} = 3$$

$$\delta = \tan^{-1} (3)$$

8. **Ans (8.00)**

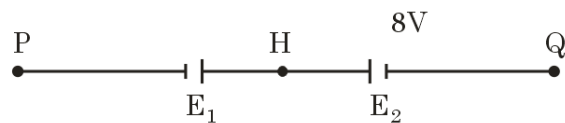
$$V_0 = \frac{V_0}{2} = \frac{V_0}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$X_L = 100 \times 4 = 400$$

$$X_C = \frac{1}{\omega C} = \frac{10^6}{100 \times 20} = 500$$

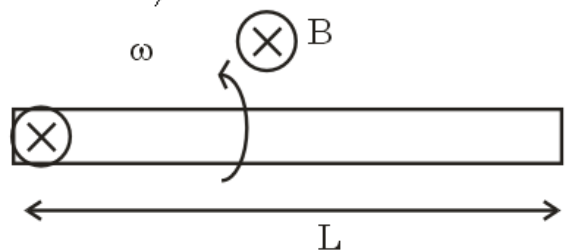
$$S = \frac{V_0}{\sqrt{100^2 + 100^2}} = \frac{V_0}{100\sqrt{2}} \Rightarrow V_0 = 500\sqrt{2}$$

9. Ans (6.00)



$$E_2 = \frac{(2)(2.4)}{2} = 8$$

$$E_1 = \frac{(2)(2)(2.4)}{2} = 2$$



$$E = \frac{Bwl^2}{2}$$

10. Ans (27.00)

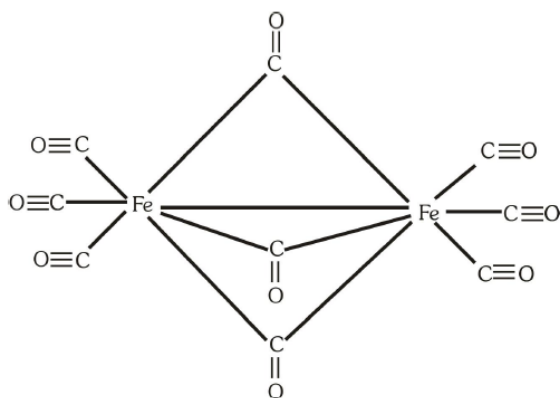
$$F = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} = \frac{1}{2\pi} \frac{1}{\sqrt{9 \times 6 \times 10^{-6}}} = \frac{10^3}{2\pi\sqrt{54}}$$

PART-2 : CHEMISTRY

SECTION-I

5. Ans (A)

Since in each metal carbonyl, EAN rule is strictly followed



11. Ans (B)

Temporary hardness is due to $\text{Ca}(\text{HCO}_3)_2$ and $\text{Mg}(\text{HCO}_3)_2$

$$n_{\text{eq.}} \text{CaCO}_3 = n_{\text{eq.}} \text{Ca}(\text{HCO}_3)_2 + n_{\text{eq.}} \text{Mg}(\text{HCO}_3)_2$$

$$\frac{w}{100} \times 2 = \frac{324 \times 10^{-3}}{162} \times 2 + \frac{146 \times 10^{-3}}{146} \times 2$$

$$\frac{w}{100} \times 2 = 4 \times 10^{-3} + 2 \times 10^{-3}$$

$$w = \frac{6 \times 10^{-3} \times 100}{2}$$

$$w = 300 \times 10^{-3} \text{ gm}$$

$$\therefore \text{Temporary hardness} = \frac{300 \times 10^{-3}}{1000} \times 10^6$$

$$= 300 \text{ ppm}$$

Permanent hardness

$$n_{\text{eq.}} \text{CaCO}_3 = n_{\text{eq.}} \text{MgCl}_2 + n_{\text{eq.}} \text{CaSO}_4$$

$$\frac{w}{100} \times 2 = \frac{190 \times 10^{-3}}{95} \times 2 + \frac{272 \times 10^{-3}}{136} \times 2$$

$$w = 400 \times 10^{-3} \text{ gm}$$

$$\therefore \text{Permanent hardness} = \frac{400 \times 10^{-3}}{1000} \times 10^6$$

$$= 400 \text{ ppm}$$

12. Ans (C)



As number of ions increases, conductance increases.

Slight decrease initially due to some dissociated CH_3COOH . After equivalence point, in place of CH_3COO^- ion, number of OH increases hence slope becomes greater.

13. Ans (B)

$$\lambda_{\text{eq.}}^{\circ}(\text{NH}_4\text{OH}) = \lambda_{\text{eq.}}^{\circ}(\text{NH}_4\text{Cl}) + \lambda_{\text{eq.}}^{\circ}(\text{NH}_4\text{OH}) - \lambda_{\text{eq.}}^{\circ}(\text{NaCl})$$

$$= 129.8 + 217.4 - 108.9$$

$$= 238.3 \text{ Ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\text{Now } \alpha = \frac{\lambda_{\text{eq.}}}{\lambda_{\text{eq.}}^{\circ}} = \frac{28.596}{238.3} = 0.12$$

$$= 12 \%$$

14. Ans (C)

Let $\text{NaHCO}_3 = a$ mole

$\text{Na}_2\text{CO}_3 = b$ mole

In the presence of phenolphthalein indicator

neg. $\text{HCl} = \text{Na}_2\text{CO}_3$

$$\therefore \frac{x \times N}{1000} = b \times 1 \quad \dots\dots(i)$$

In presence of methyl orange

neg. $\text{HCl} = \text{neg. NaHCO}_3 (\text{original}) + \text{neg.}$

$\text{NaHCO}_3 (\text{formed})$

$$\therefore \frac{y \times N}{1000} = a \times 1 + b \times 1 \quad \dots\dots(ii)$$

From equation (i) and (ii)

We get, V_{HCl} for original $\text{NaHCO}_3 = (y - x)$ ml.

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$\begin{aligned} & \frac{4(\sin 9^\circ \sin 51^\circ \sin 69^\circ)(\sin 21^\circ \sin 39^\circ \sin 81^\circ)}{\sin 54^\circ} \\ &= \frac{4 \times \frac{1}{4} \sin 27^\circ \times \frac{1}{4} \sin 63^\circ}{\sin 54^\circ} \\ &= \frac{1}{8} \frac{2 \sin 27^\circ \sin 63^\circ}{\sin 54^\circ} = \frac{1}{8} \frac{(\cos 36^\circ - \cos 90^\circ)}{\sin 54^\circ} = \frac{1}{8} \end{aligned}$$

2. Ans (D)

$$(y + 2)^2 = -4x - 2 + 4$$

$$(y + 2)^2 = -4 \left(x - \frac{1}{2} \right)$$

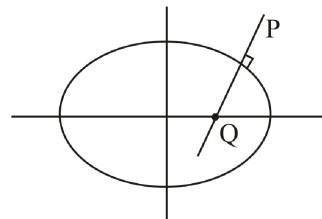
$$\Rightarrow \text{Directrix is } x - \frac{1}{2} = 1$$

$$x = 3/2$$

3. Ans (C)

$P(a \cos \theta, b \sin \theta)$

$Q(e^2 a \cos \theta, 0)$



$$\begin{aligned} PQ^2 &= [a \cos \theta (e^2 - 1)]^2 + (b \sin \theta)^2 \\ &= a^2 \cos^2 \theta (e^2 - 1)^2 + a^2 (1 - e^2) \sin^2 \theta \\ &= a^2 (e^2 - 1) [\cos^2 \theta (e^2 - 1) - \sin^2 \theta] \\ &= a^2 (e^2 - 1) (e^2 \cos^2 \theta - 1) \\ &= a^2 (1 - e^2) (1 - e^2 \cos^2 \theta) \\ &= (1 - e^2) (a - ae \cos \theta) (a + ae \cos \theta) \\ &= (1 - e^2) r r_1 = \frac{b^2}{a^2} r r_1 \end{aligned}$$

4. Ans (C)

$$y(x - 2) = 1, \quad \sqrt{x} + \sqrt{y - 2} = \sqrt{x + y};$$

$$x > 2, y \geq 2$$

$$\text{Now, } x + y - 2 + 2\sqrt{x(y - 2)} = x + y$$

$$\Rightarrow x(y - 2) = 1 \Rightarrow y \left(\frac{1}{y - 2} - 2 \right) = 1$$

$$\Rightarrow y(1 - 2y + 4) = y - 2$$

$$\Rightarrow y^2 - 2y - 1 = 0$$

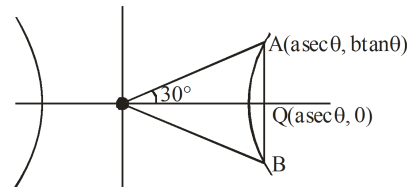
$$\Rightarrow y = \frac{2 \pm 2\sqrt{2}}{2} = 1 \pm \sqrt{2}$$

$$\Rightarrow y = 1 + \sqrt{2}$$

$$\Rightarrow x - 2 = \sqrt{2} - 1 \Rightarrow x = \sqrt{2} + 1$$

$$\Rightarrow x + y = 2(\sqrt{2} + 1)$$

5. Ans (D)



$$\tan 30^\circ = \frac{b \tan \theta}{a \sec \theta}$$

$$\Rightarrow \frac{a}{\sqrt{3} b} = \sin \theta$$

$$\Rightarrow \frac{a^2}{3b^2} < 1$$

$$\Rightarrow e^2 - 1 > \frac{1}{3} \Rightarrow e > \frac{2}{\sqrt{3}}$$

6. **Ans (C)**

$$2^{\sin^2 x} + 4 \cdot 2^{\cos^2 x} = 6; 2^{\sin^2 x} = t, \quad t \in [1, 2]$$

$$\Rightarrow t^2 - 6t + 8 = 0 \Rightarrow t = 2 \text{ or } 4$$

$$\Rightarrow t = 2 \Rightarrow \sin^2 x = 1 \Rightarrow 4 \text{ solutions.}$$

7. **Ans (A)**

$$\frac{\cot \alpha_1}{\cot \alpha_2} = \frac{t_1}{t_2} = 2; a t_1 t_2 = h, a(t_1 + t_2) = k$$

$$\Rightarrow a t_1 \cdot \frac{t_1}{2} = h \Rightarrow t_1^2 = \frac{2h}{a}, t_2^2 = \frac{h}{2a}$$

$$\Rightarrow a^2 \left(\frac{2h}{a} + \frac{h}{2a} + \frac{2h}{a} \right) = k^2$$

$$\Rightarrow \frac{9ah}{2} = k^2$$

$$\Rightarrow 2y^2 = 9ax$$

8. **Ans (C)**

$$2 \log_{\frac{x}{2}} x + 4 \log_{4x} x = 9 \log_{2x} x \quad (x > 0)$$

one solution is $x = 1$,

$$\text{Let } \log_x 2 = t$$

$$\text{So, } \frac{2}{1-t} + \frac{4}{2t+1} = \frac{9}{t+1}$$

Now solve

9. **Ans (C)**

$$\frac{x}{a} \cos \left(\frac{\alpha + \beta}{2} \right) + \frac{y}{b} \sin \left(\frac{\alpha + \beta}{2} \right) = \cos \left(\frac{\alpha - \beta}{2} \right)$$

$$\ell x + my = -n; |\alpha - \beta| = \pi/2$$

$$\frac{\cos \left(\frac{\alpha + \beta}{2} \right)}{a\ell} = \frac{\sin \left(\frac{\alpha + \beta}{2} \right)}{bm} = \frac{-1}{\sqrt{2}n}$$

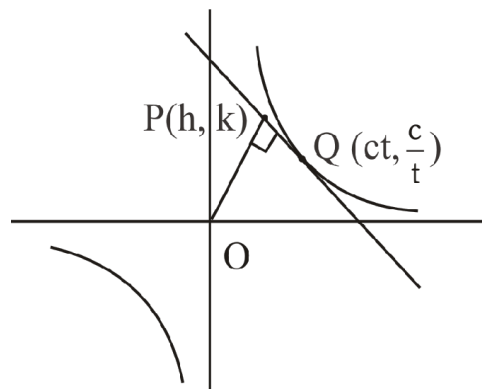
$$\Rightarrow \cos \left(\frac{\alpha + \beta}{2} \right) = -\frac{a\ell}{\sqrt{2}n}, \sin \left(\frac{\alpha + \beta}{2} \right) = \frac{-bm}{\sqrt{2}n}$$

$$\Rightarrow a^2 \ell^2 + b^2 m^2 = 2n^2$$

10. **Ans (D)**

Equation of tangent

$$x + yt^2 = 2ct$$



$$m_{op} = \frac{k}{h} \Rightarrow \text{Slope of tangent} = \frac{-h}{k}$$

$$\Rightarrow -\frac{1}{t^2} = -\frac{h}{k} \Rightarrow t^2 = \frac{k}{h}$$

$$\text{So, } h + k \frac{k}{h} = 2c \sqrt{\frac{k}{h}} \Rightarrow h^2 + k^2 = 2c \sqrt{hk}$$

$$\Rightarrow (x^2 + y^2)^2 = 4c^2 xy$$

11. **Ans (D)**

$$\sin^2 A + \sin^2 B = \sin C = \sin(A + B)$$

$$\Rightarrow \sin A \sin A + \sin B \sin B = \sin A \cos B + \sin B \cos A$$

$$\text{Now, } A + B < \frac{\pi}{2}$$

$$\Rightarrow A < \frac{\pi}{2} - B$$

$$\Rightarrow \sin A < \cos B$$

$$\text{and } \cos A > \sin B$$

$$\text{Hence, L.H.S} < \text{R.H.S}$$

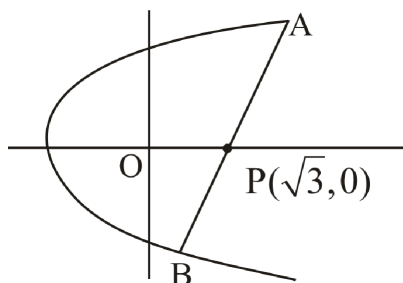
$$\text{Therefore } A + B = \frac{\pi}{2} \Rightarrow C = \frac{\pi}{2}$$

12. **Ans (A)**

Equation of line can be written as

$$\frac{x - \sqrt{3}}{\cos \frac{\pi}{3}} = \frac{y - 0}{\sin \frac{\pi}{3}} = r$$

$\therefore$ Point = $\left(\sqrt{3} + \frac{r}{2}, \frac{r\sqrt{3}}{2}\right)$ lies on parabola.



$$\therefore \frac{3}{4}r^2 = \sqrt{3} + \frac{r}{2} + 2$$

$$\Rightarrow 3r^2 - 2r - 4(2 + \sqrt{3}) = 0$$

$$\therefore PA \cdot PB = r_1 r_2 = \left| \frac{-4(2 + \sqrt{3})}{3} \right| = \frac{4(2 + \sqrt{3})}{3}$$

13. **Ans (A)**

The asymptotes are $y = \pm \frac{\sqrt{3}}{2}x$ and the extremities of double ordinate be

$$P \left(h, \frac{\sqrt{3}}{2} \sqrt{h^2 - 4} \right) \text{ and } P' \left(h, -\frac{3}{2} \sqrt{h^2 - 4} \right)$$

$$\Rightarrow (PQ)(PQ') = 3$$

14. **Ans (C)**

$$\log_{10} xy = 2$$

$$xy = 100$$

$$\frac{x + y}{2} \geq \sqrt{xy}$$

$$(x + y) \geq 20$$

15. **Ans (A)**

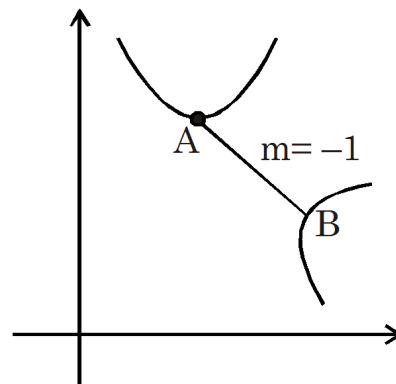
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1; a > b > 0$$

16. **Ans (D)**

Consider parabolas

$$x^2 - 8x - 4y + 40 = 0 \text{ and } y^2 - 4x - 8y + 40 = 0$$

$$\Rightarrow (x - 4)^2 = 4(y - 6) \text{ and } (y - 4)^2 = 4(x - 6)$$



$$A = (6, 7); B = (7, 6)$$

$$\Rightarrow AB = \sqrt{2}$$

17. **Ans (B)**

$$\sum \cos 3P = 1$$

$$\Rightarrow \Pi \sin \frac{3P}{2} = 0$$

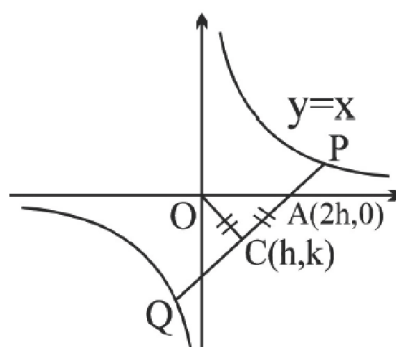
$$\Rightarrow P = \frac{2\pi}{3}$$

$$\text{or } Q = \frac{2\pi}{3} \text{ or } R = \frac{2\pi}{3}$$

18. **Ans (B)**

Chord with a given middle point is $yh + xk = 2hk$

$$\Rightarrow \frac{x}{h} + \frac{y}{k} = 2$$



So, by observation OCA is isosceles with $OC = CA$.

19. Ans (B)

Given equation of ellipse as $x^2 + 4y^2 = 4$ (1)

and equation of circle as $x^2 + y^2 = 2$ (2)

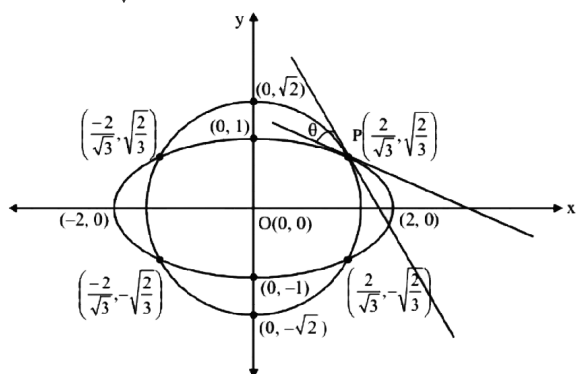
∴ Equation (1) – Equation (2)

$$\Rightarrow 3y^2 = 2 \Rightarrow y = \pm \sqrt{\frac{2}{3}}$$

$$\text{Also, } x^2 = 2 - \frac{2}{3}$$

$$\Rightarrow x^2 = \frac{4}{3}$$

$$\Rightarrow x = \pm \frac{2}{\sqrt{3}}$$



Let m_1 = slope of tangent at $P \left(\frac{2}{\sqrt{3}}, \sqrt{\frac{2}{3}} \right)$ on ellipse

$$\Rightarrow m_1 = \frac{-1}{2\sqrt{2}}$$

Also M_2 = slope of tangent at $P \left(\frac{2}{\sqrt{3}}, \sqrt{\frac{2}{3}} \right)$ on circle

$$\Rightarrow m_2 = -\sqrt{2}$$

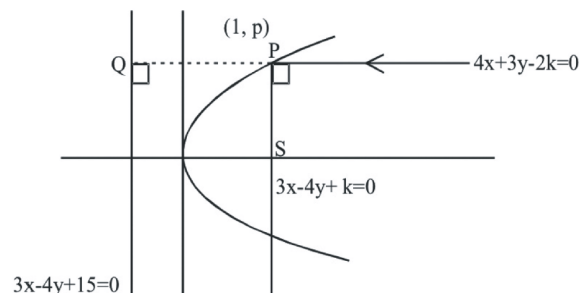
Let θ be the acute angle between tangents at

$P \left(\frac{2}{\sqrt{3}}, \sqrt{\frac{2}{3}} \right)$ on

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\frac{-1}{2\sqrt{2}} + \sqrt{2}}{1 + \left(\frac{1}{2\sqrt{2}} \right) \cdot \sqrt{2}} \right|$$

$$= \left| \frac{\frac{3}{2\sqrt{2}}}{\frac{3}{2}} \right| = \frac{1}{\sqrt{2}}$$

20. Ans (A)



$(1, p)$ satisfies

$$4x + 3y - 2k = 0 \text{ \& } 3x - 4y + k = 0$$

$$\therefore p = 2 \text{ \& } k = 5$$

now $P \equiv (1, 2)$

$$\text{Now } PS = PQ = 2$$

$$\therefore LR = 4$$

PART-3 : MATHEMATICS

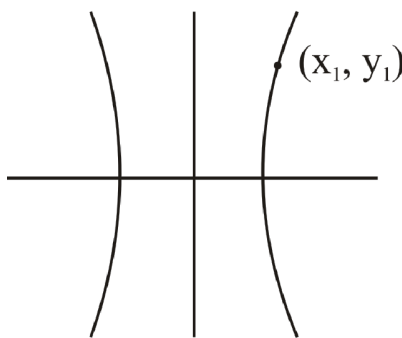
SECTION-II

1. Ans (0.00)

$$x^2 - y^2 = a^2$$

eqⁿ of Tangent is $xx_1 - yy_1 = a^2$

and, eq. of normal $\frac{x}{x_1} + \frac{y}{y_1} = 2$



$$a_1 = \frac{a^2}{x_1}, a_2 = 2x_1 \Rightarrow a_1 a_2 = 2a^2$$

$$\text{Also, } b_1 = \frac{-a^2}{y_1}, b_2 = 2y_1 \Rightarrow b_1 b_2 = -2a^2$$

2. Ans (17.00)

$$\log_2(\log_2(\log_3 x)) = \log_2 \log_3(\log_2 x) = 0$$

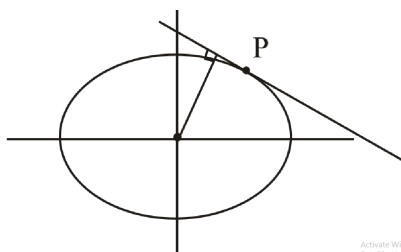
$$\Rightarrow x = 3^2 = 9, y = 2^3 = 8$$

$$\Rightarrow x + y = 17$$

3. **Ans (4.00)**

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

$$bx \cos \theta + ay \sin \theta = ab$$



$$d = \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$\Rightarrow d^2 = \frac{a^2 b^2}{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$(PF_1 - PF_2)^2 = (PF_1 + PF_2)^2 - 4PF_1 \cdot PF_2$$

$$= 4a^2 - 4(a^2 - a^2 e^2 \cos^2 \theta)$$

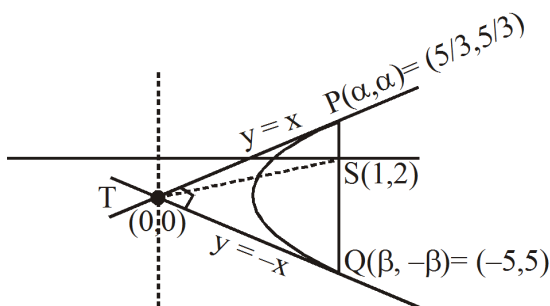
$$m = \frac{4a^2 e^2 \cos^2 \theta}{a^2 \left(1 - \frac{a^2 \sin^2 \theta + b^2 \cos^2 \theta}{a^2} \right)}$$

$$= \frac{4e^2 \cos^2 \theta a^2}{a^2 \cos^2 \theta - b^2 \cos^2 \theta} = 4$$

4. **Ans (11.00)**

(0, 0) lies on directrix

$$m_{ST} \times m_{SP} = -1$$



$$2 \times \frac{\alpha - 2}{\alpha - 1} = -1 \Rightarrow 2\alpha - 4 = 1 - \alpha$$

$$\Rightarrow 3\alpha = 5 \Rightarrow \alpha = 5/3$$

Also,

$$-\frac{\beta - 2}{\beta - 1} \times 2 = -1 \Rightarrow 2\beta + 4 = \beta - 1 \Rightarrow \beta = -5$$

$$\therefore \text{Mid pt of PQ is R} \therefore R = \left(\frac{-5}{3}, \frac{10}{3} \right)$$

$$\therefore \text{Slope of RT} = -2 \Rightarrow \text{Slope of directrix} = \frac{1}{2}$$

$$\therefore \text{eq. of Directrix : } x - 2y = 0$$

$$\therefore LR = 2 \left| \frac{1 - 4}{\sqrt{5}} \right| = \frac{6}{\sqrt{5}}$$

5. **Ans (3.25)**

$$(1 + \sin t)(1 + \cos t) = 5/4$$

$$\sin t + \cos t + \sin t \cos t = \frac{1}{4} \quad \text{--- (1)}$$

$$1 - \sin t - \cos t + \sin t \cos t = y$$

$$\sin t + \cos t - \sin t \cos t = 1 - y \quad \text{--- (2)}$$

eliminate t from (1) and (2)

6. **Ans (2.00)**

Let $\left(t, \frac{1}{t} \right)$ be any point on $xy = 1$

$$\Rightarrow xy' + y = 0$$

$$\Rightarrow y' = \frac{-y}{x}$$

$$\Rightarrow y' = -\frac{y}{t^2}$$

$$\Rightarrow \frac{-a}{b} = t^2$$

$\Rightarrow a$ & b are of opp. sign.

7. **Ans (4.00)**

$$|x - 3|^{\left(\frac{x+1}{4} \right)} = |x - 3|^{\left(\frac{x-2}{3} \right)}$$

$$\Rightarrow x - 3 = 0, 1, -1$$

$$\text{or } \frac{x+1}{4} = \frac{x-2}{3}$$

8. **Ans (21.00)**

$$25 \left(\frac{3x + xy - 7}{5} \right)^2 + 9 \left(\frac{9x - 3y - 1}{5} \right)^2 = 225$$

$$\text{Major axis } 3x + 4y = 7$$

$$\text{Minor axis } 4x - 3y = 1$$

$$\text{Under (iii) } e = 4/5$$

$$\text{Foci} = \left(-\frac{11}{5}, \frac{17}{5} \right), \left(\frac{21}{5}, \frac{-7}{5} \right)$$

$$\text{Latus Rectum } 4x - 3y = 21, 4x - 3y = -19$$

9. **Ans (3.00)**

$$\sum_{r=1}^n \frac{\sin(2^r - 2^{r-1})}{\cos 2^r \cos 2^{r-1}} = \sum_{r=1}^n (\tan 2^r - \tan 2^{r-1}) = \tan 2^n - \tan 1$$

10. **Ans (7.00)**

Let $P(t_1)$ and $Q(t_2)$ are point of $y^2 = 8x$

$$\text{Now, } 2t_1^2 + 2t_2^2 = 17 \text{ and } (2t_1^2)(2t_2^2) = 11$$

$$ST^2 = SP \cdot SQ = 2(1 + t_1^2)2(1 + t_2^2) = 34 + 4 + 11$$

$$ST = \sqrt{49}$$